Manan Gandhi

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Backend & Systems Engineer • Software Security • Open Source • 8x Hackathon Winner

Experience

IkiQuant Technologies — Remote, India

Mid-Frequency Trading Intern - US Index Options — *Mar 2026 – Present*

- Built the real-time **risk management system** supporting live trading operations and continuous exposure monitoring.
- Reworked critical execution and trade-processing paths, reducing strategy execution latency by **15× across all strategies**.
- Optimized market-data and Options Greeks pipelines by **3×**, improving real-time computation throughput for live strategy execution.
- Built **real-time monitoring, research, and live market-data dashboards**, including interactive charting similar to TradingView.

Sykes & Rays Equities — Mumbai, India

Software Engineering Intern — *May 2025 – Jul 2025*

- Developed automated trading workflows for strategy execution using Zerodha Kite API and OpenAlgo.
- Resolved a critical issue in the OpenAlgo Zerodha Connector, improving reliability and earning public maintainer recognition.
- Automated post-market data workflows, reducing traders' manual effort by **90%**.

Projects

IntelliAnnounce — Website

Real-Time Market Intelligence Platform

- Built a **speech-first market intelligence platform** used regularly by traders to monitor company disclosures without continuously reading the screen.
- Architected the data pipeline with **Redis as the primary hot-path datastore** for live announcements and nightly cron-based persistence to PostgreSQL, minimizing ingestion latency and enabling high-throughput reads during market hours.
- Designed personalized watchlists, intelligent filtering, real-time alerts, and AI-generated summaries to surface relevant market-moving information and reduce information overload.

Sandboxed Online Code Judge — GitHub

Sandboxed Code Execution Platform

- Built an online coding platform used by **200+ users**, supporting submissions, leaderboards, editorials, and AI-assisted hints.
- Designed an asynchronous execution pipeline using **Docker containers for isolation and Redis Pub/Sub messaging** for Python, C, C++, and Java submissions with controlled resource usage.
- Hardened execution with hidden test cases, resource limits, and persistent submission telemetry for reliable multi-user workloads.

OpenQuant — GitHub

Options Analytics and Strategy Research Platform

- Built a **NIFTY options analytics platform** for open interest, volume, put-call ratios, volatility, Greeks, and strike-level market analysis.
- Developed tools for **strategy construction, payoff analysis, historical backtesting, and anomaly detection** to support options research and experimentation.
- Added AI-assisted market interpretation and strategy insights; won **1st place at FOSS Club MPSTME CodeForge 2026**.

Adeon — GitHub

AI-Powered Software Development Platform

- Built a developer platform for **code review, vulnerability analysis, documentation, testing, and development environment provisioning**.
- Designed a multi-service architecture separating analysis and automation workflows, making platform capabilities easier to extend independently.
- Ranked **Top 35 globally** among **1,500+ projects** at the 100X Engineers Buildathon.

FluxGate — GitHub

Self-Hosted Continuous Deployment Platform

- Built a self-hosted deployment platform that automatically rebuilds and deploys applications when code changes are pushed to GitHub.
- Added isolated deployments, branch-aware environments, real-time build logs, failure alerts, and multi-environment deployment support.
- Implemented **natural-language infrastructure management** to simplify deployment configuration and repository setup.

Achievements

- **8× Hackathon Winner** across national and international competitions, including the Supabase Launch Week Hackathon.
- Responsibly disclosed multiple **high-severity security vulnerabilities** in production systems including Matiks, Unstop, and iicpc.com.
- Publicly recognized by OpenAlgo maintainer Rajandran R for a critical Zerodha Connector fix (LinkedIn, X).

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL, Java, Dart, C++, Kotlin, Bash

Systems & Data: Docker, Redis, Apache Kafka, ClickHouse, Linux, AWS, Azure, PostgreSQL, Apache Spark

Backend: Node.js, Express.js, FastAPI, Django, Flask

Frontend / Mobile: React.js, Next.js, Flutter, React Native, Jetpack Compose

Education

Mukesh Patel School of Technology Management and Engineering — Mumbai, India

B.Tech in Computer Engineering — *2022 – 2028*

Kapol Vidyanidhi International School — Mumbai, India

ICSE — *Graduated: 2022*